

BENZAPEN 2.4 miu Injection
Sterile Penicillin G Benzathine 1.8 g (2,400,000 Units)

Product Name

BENZAPEN 2.4 miu Injection

Product Description

A white powder (for injection) in glass vials.

Composition

Each vial contains Sterile Penicillin G Benzathine 1.8 g (2,400,000 Units).

Pharmacodynamic

Benzathine penicillin is a depot preparation of benzylpenicillin with the same mode of action as that benzylpenicillin. The penicillin nucleus consist of thiazolidine ring connected to β -lactam ring to which is attached a side chain. The β -lactam ring plays an important role in the antibacterial activity of β -lactam antibiotics. It can be opened by β -lactamase produced by enzyme-producing organisms. As soon as the β -lactam ring is opened by the β -lactamase the hydrolysed penicillin is inactivated. The side-chain determines most of the pharmacological properties as well as the characteristics of the antimicrobial activity penicillins. Penicillin kills bacteria by interfering with the synthesis of the bacteria cell wall. This is composed of peptidoglycan which is a heteropolymeric structure that provides the cell wall with mechanical stability. The final stage in the synthesis of peptidoglycan involves the completion of the cross-linking between the terminal glycine residue of the pentaglycine bridge and the fourth residue of the pentapeptide (d-alanine). The transpeptidase that performs this step is inhibited by penicillins and cephalosporins, the β -lactam ring of which acts as an analogue of acyl-d-ananyl-d-alanine. As a result the bacteria cell wall is weakened and then ruptures. Recently it was proved that there are penicillin-binding-proteins (PBPs) in the cell envelope of bacteria which are the targets for β -lactam antibiotics.

Pharmacokinetics

Absorption and serum levels

Benzathine penicillin is stable in the presence of gastric juice, but not absorption from the gastro-intestinal tract. Benzathine penicillin is very slowly absorbed from intramuscular depot. It is hydrolysed to benzylpenicillin and produces sustained low levels of penicillin in the blood for an average 4 weeks period. A dose of 1.2 million units given intramuscularly produces a concentration in plasma of 150u/ltr on the first and 30u/ltr on the fourteenth day. After a single injection of 2.4 million units of benzathine penicillin in young subjects, the mean serum penicillin concentration was 340u/ltr after hours.

Distribution

Approximately 60% of benzylpenicillin is bond to serum proteins. The drug is distributed throughout the body tissues in widely varying amounts. Highest levels are found in kidneys with lesser amounts in the liver skin and in intestines. Benzypenicillin penetrates into all other tissues and spinal fluid to a lesser degree. With normal kidney function, the

drug is excreted rapidly by tubular excretion. In neonates and young infants and in individuals with impaired kidney function, excretion is completely delayed.

Metabolism of excretion

Benzathine penicillin is partially metabolised to inactive penicilloic acid in the liver in the same manner as is benzylpenicillin. Once benzylpenicillin is released from repository forms, it is excreted by the kidney. About 70 to 80% of injected penicillin can be recovered in the urine and about 20% are excreted by the biliary system into gastrointestinal tract. Because absorption in to the blood from injection site is continued over a long period, the excretion of active penicillin in the urine is prolonged. Studies have revealed urinary excretion persisting from 84 days to more than 100 days. It seems reasonable that benzathine penicillin persists for a longer time in tissues than in the blood stream.

Indications

Sterile Pencillin G Benzathine is indicated in the treatment and prophylaxis of infections caused by organisms with high penicillin sensitivity:

Sterile Pencillin G Benzathine is used as a single agent in the treatment of

- Acute tonsillitis
- Scarlet fever
- Erysipela (chronic), erysipeloid
- Wound infections, bite wounds
- Syphilis (a single dose is sufficient in primary syphilis)
- Other treponemal infections (yaws, bejel, pinta)

Prophylaxis

Sterile Pencillin G Benzathine is used as a single agent in the prevention of

- Rheumatic fever (chorea, rheumatic heart disease)
- Poststreptococcal glomerulonephritis
- Scarlet fever (in exposed contacts)
- Erysipela
- Syphilis (in exposed contacts)

Recommended Dosage

Sterile Penicillin G Benzathine is administered by intramuscular injection.

Treatment of syphilis

1) Congenital syphilis in neonates or infants ('without neurological involvement,):

1, 200 000 units injected at 2 sites.

2) Primary, secondary, latent syphilis,

- a) Primary : I dose of 2, 400 000 units injected at 2 sites.
- b) Secondary : 2 doses of Sterile pencillin G benzathine 2.4 MU Injected at 2 sites.
Repeat treatment if clinical symptoms recur or if laboratory tests are persistently positive.

- c) Late (tertiary, seropositive): 1-2 doses of Sterile penicillin G benzathine 2.4 MU weekly for 3 to 5 weeks.

Treatment of yaws

Children under 12 years: 600 000 units.

Adults : 1, 200 000 units to 2, 400 000 units

Contacts and persons with latent infections are given half of these doses.

Treatment of pinta

Children : 1, 200 000 units

Adults : 2, 400 000 units

Other infections

Children under 12 years : 600 000 units at intervals of 3 days or daily, depending on the severity of the disease. Alternatively, 1, 200 000 units at intervals of 2 to 4 weeks.

Adults : 1, 200 000 units or 2, 400 000 units once a week.

Prophylaxis

Prevention of recurrent rheumatic fever, rheumatic endocarditis, chorea, poststreptococcal glomerulonephritis and erysipela:

Children under 12 years : 600 000 units at intervals of 14 days or 1, 200 000 units at intervals of 4 weeks.

Adults 1, 200 000 units to 2, 400 000 units at intervals of 4 weeks.

Prevention of scarlet fever in contacts:

Children under 12 years : 600 000 units or, alternatively, 1, 200 000 units at weekly intervals.

Adults : 2, 400 000 units

To prevent complications, patients with streptococcal disease should receive treatment for at least 10 days.

This can generally be ensured with a single injection of 600 000 units in children under age 12 years, with a single injection of 1, 200 000 units in children over age 12 years and with a single injection of 2, 400 000 units in adults.

Do not apply more than 50 mg of PVP (equivalent to 2, 400 000 units) at one injection site. Inject higher single doses (e. g. 2 doses of 2, 400 000 units) at 2 different sites.

How to use

Dissolve the contents in 8ml or more of Sterilized Water for Injections immediately before use.

Incompatibility

Extemporaneous admixtures of beta-lactam antibacterials and aminoglycosides may result in substantial mutual inactivation. If these groups of antibacterials are administered concurrently, they should be administered in separate sites at least 1 hour apart. Do not mix them in the same intravenous bag, bottle, or tubing.

When aminoglycosides and penicillins are administered separately by different routes, a reduction in aminoglycosides serum concentration may occur. Usually this is clinically significant only in patients with severely impaired renal function when the excretion of both medications is delayed.

Contraindications

History of hypersensitivity to penicillins and/or other ingredients. In patients known to be hypersensitive to cephalosporins, potential cross allergenicity should be kept in mind. In diseases like severe pneumonia, empyema, sepsis, pericarditis, meningitis, peritonitis or arthritis requiring high penicillin serum levels, treatment should be started with the water-soluble alkali salt of benzylpenicillin. If neurological involvement cannot be ruled out in patients with congenital syphilis, penicillin dosage forms producing higher CSF concentrations should be used. Special caution should be exercised in patients with an allergic diathesis or with bronchial asthma.

Warnings and Precautions

Patients should be alerted to the potential occurrence of allergic reactions and instructed to report them.

Particular caution should be exercised in patients with allergic diathesis or with bronchial asthma.

Patients should be watched for 30 minutes after drug administration and adrenalin should be kept ready for injection.

If allergic reactions occur, the drug should be withdrawn and the usual treatment with adrenalin, antihistamines and corticosteroids should be instituted.

To suppress or mitigate Jarisch-Herxheimer reactions (see 'Side effects') prednisolone, 50 mg. or an equivalent steroid, may be administered at the time of the first antibiotic dose. In patients with cardiovascular or meningovascular syphilis Jarisch-Herxheimer reactions can be prevented by prednisolone, 50 mg daily, or an equivalent steroid, for 1 to 2 weeks. In diabetics delayed absorption from the intramuscular depot should be remembered. During long-term treatment the blood count and renal function should be monitored. The potential overgrowth of resistant organisms should be kept in mind during long-term treatment. Patients developing secondary infections should be treated by suitable measures.

In patients treated for venereal diseases who are suspected of having co-existent syphilis dark-field examinations should be ordered prior to treatment and serologic tests should be obtained for at least 4 months. In patients with congenital syphilis the CSF should be examined prior to treatment.

Sustained severe diarrhea should prompt suspicion of antibiotic related pseudomembranous colitis (blood-streaked, mucoid, watery diarrhea dull, diffuse to colicky abdominal pain; fever and occasionally also tenesms). As this condition may be life-threatening, Sterile Penicillin G Benzathine should immediately be withdrawn and

treatment guided by bacteriological studies (e. g. with oral vancomycin, 250 mg q. i. d.) should be instituted. Antiperistaltics are contraindicated.

To avoid sciatic nerve damage, infants and small children should not be injected into the upper outer quadrant of the buttock except in special cases, e. g. in the presence of extensive burns. Sterile Penicillin G Benzathine must not be injected subcutaneously, intravenously or intrathecally or instilled into body cavities.

Inadvertent subcutaneous injection may cause painful indurations. Pain is relieved by ice packs.

Inadvertent intravascular injection may cause Hoigne's syndrome (shock symptoms with a feeling of impending doom, confusion, hallucinations, at times also cyanosis, tachycardia and motor abnormalities, but no circulatory collapse). Attributable to microembolism of the suspension, these manifestations spontaneously disappear within 1 hour. In severe cases patients should be given sedatives by injection.

Particularly in children, inadvertent intra-arterial injection may cause severe complications like vessel occlusion, thrombosis and gangrene. These are heralded by pale patches on the skin of the buttocks. If injected at high pressure, the injected fluid may be pushed upstream into the common iliac arteries, the aorta or the spinal arteries.

Interactions with Other Medicaments

As penicillins are only active against proliferating micro-organisms, Sterile Penicillin G Benzathine should not be combined with bacteriostatic antibiotics. Combinations with other antibiotics should only be considered, if their effects can be expected to be synergistic or at least additive. Except if a synergistic action is proven, the combination partners should be fully dosed.

Competitive inhibition of drug elimination should be remembered, whenever antiinflammatories, antirheumatics, antipyretics (particularly indomethacin, phenylbutazone and salicylates in high doses) or probenecid are used concomitantly. Like other antibiotics, Sterile Penicillin G Benzathine may occasionally reduce the efficacy of oral contraceptives.

To preclude undesirable chemical reactions, administration with mixed-content syringes should be avoided.

Pregnancy and Lactation

There is currently no evidence suggesting that Sterile Penicillin G Benzathine, if administered during pregnancy, has an embryotoxic, teratogenic or mutagenic effect. It should, however, be remembered that benzathine benzylpenicillin may be present in breast milk

Side Effects

Hypersensitivity reactions

Allergic reactions are uncommon. Potential allergic reactions include urticaria, angioneurotic edema, erythema multiforme, exfoliative dermatitis, fever, joint pain or

anaphylactic shock with a collapse and anaphylactoid reactions (asthma, purpura, gastrointestinal symptoms).

Gastro-intestinal side effects

Stomatitis and glossitis are, at times, seen. In patients developing diarrhea during treatment pseudomembranous colitis should be thought of.

Hematological side effects

Positivity of the direct Coombs' test, hemolytic anemia, leucopenia, thrombocytopenia and agranulocytosis have been reported, but are extremely rare.

Other side effects

Neuropathy, nephropathy.

Patients treated for syphilis may develop Jarisch-Herxheimer reactions secondary to bacteriolysis.

Infants may show local reactions.

Although extremely rare, accumulation of PVP in the reticuloendothelial system (RES) or local PVP accumulation and foreign body granulomas mimicking tumors cannot altogether be excluded.

Symptoms and Treatment of Overdose

Penicillin in overdosage has the potential to cause neuromuscular hyper-irritability or convulsive seizures.

Presentation

One folded paper box contains 10 vials or 50 vials.

Storage Condition

Store below 30°C.

Shelf Life

36 months from the date of manufacture if kept as recommended.

Registration number

BENZAPEN 2.4 miu Injection: MAL07061452AZ
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